1. Weighted Scoring Algorithm

- Each project receives a score based on how well it matches a student's preferences.
- The algorithm compares a student's information to each project and awards points whenever there's a match.

(you can think of it as a scoring board)

Pros

- Very easy to understand
- Easy to explain to users
- Fast
- Adjustable (change weights whenever needed)
- Doesn't require lots of historical data

Cons

- Choosing weights can be subjective.
- Doesn't automatically ensure every project receives students.
- Doesn't account for competition between students.

2. Stable Matching (Gale-Shapley Algorithm)

- algorithm used for the National Resident Matching Program (medical residency matching).
- The project & intern both rank each other, both lists are compared to one another.
- The algorithm repeatedly proposes until every student is matched with the best available project while preventing unstable matches.

Pros

- Fair
- Nobody can improve their match without making someone else worse off.
- Widely trusted

Cons

- More difficult to implement.
- Requires projects to rank students.

- Doesn't work well if only students have preferences.

3. Hungarian Algorithm

- Creates a matrix of compatibility scores.
- The algorithm finds the assignment that maximizes the total score.

(would have to first need compatibility scores; would need to use another algorithm)

Pros

- Finds the mathematically optimal assignment.
- No duplicated assignments.
- Very accurate.

Cons

- More complicated.
- Doesn't naturally incorporate ranked preferences unless converted into scores.
- Harder to explain.

4. Recommendation System (Content-Based Filtering)

- Similar to Netflix or Spotify.
- Algorithm looks for any keyword overlap b/w the application & project description.

Pros

- Personalized
- Easy to add more preferences later
- Doesn't require projects to rank students

Cons

- Recommendations aren't guaranteed matches.
- Doesn't assign projects automatically.

5. Cosine Similarity

- Very common in machine learning.
- Represent each student and project as vectors.
- tells you how alike they are in numbers.

Pros

- Fast
- Scales well
- Great with many interests

Cons

- Doesn't consider project capacity.
- Doesn't assign students automatically.

6. Multi-Criteria Decision Making (MCDM)

- Instead of one score, evaluate many categories
- Each category gets a weight.

Pros

- Flexible
- Transparent
- Easy to explain

Cons

- Weight selection can be subjective

7. Hybrid Recommendation System

- This is what many modern platforms use.
- Instead of relying on a single algorithm:
    - Filter projects based on eligibility (major, class standing, required skills).
    - Use a weighted scoring system to rank remaining projects.
    - Use similarity (e.g., cosine similarity) to personalize recommendations.

- If projects have limited capacity, use a matching algorithm (like Gale–Shapley or Hungarian) to finalize assignments.